

Patterns of Mass Gathering Events in 2024 from DigiHero Study

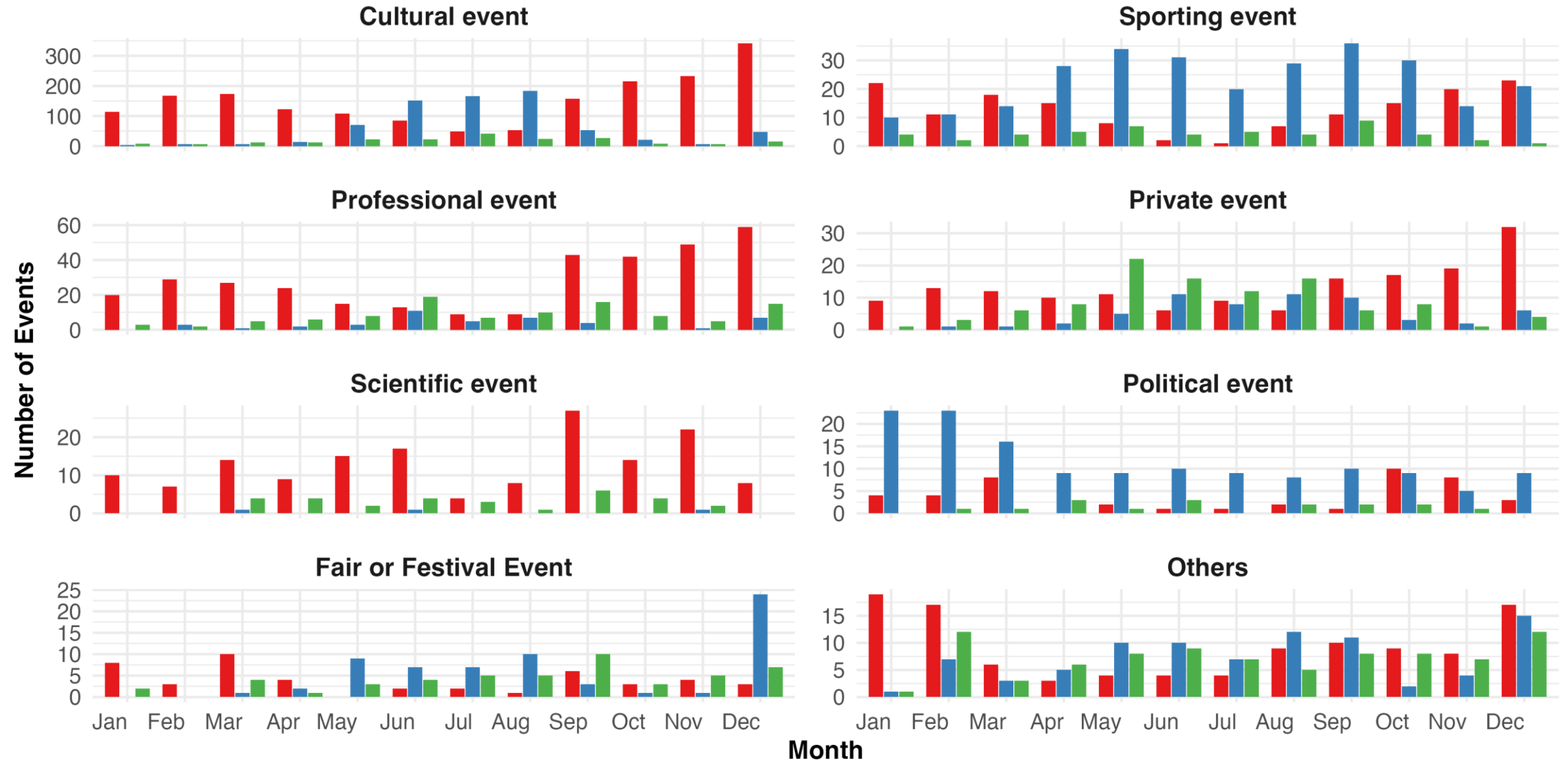
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08-12-2025

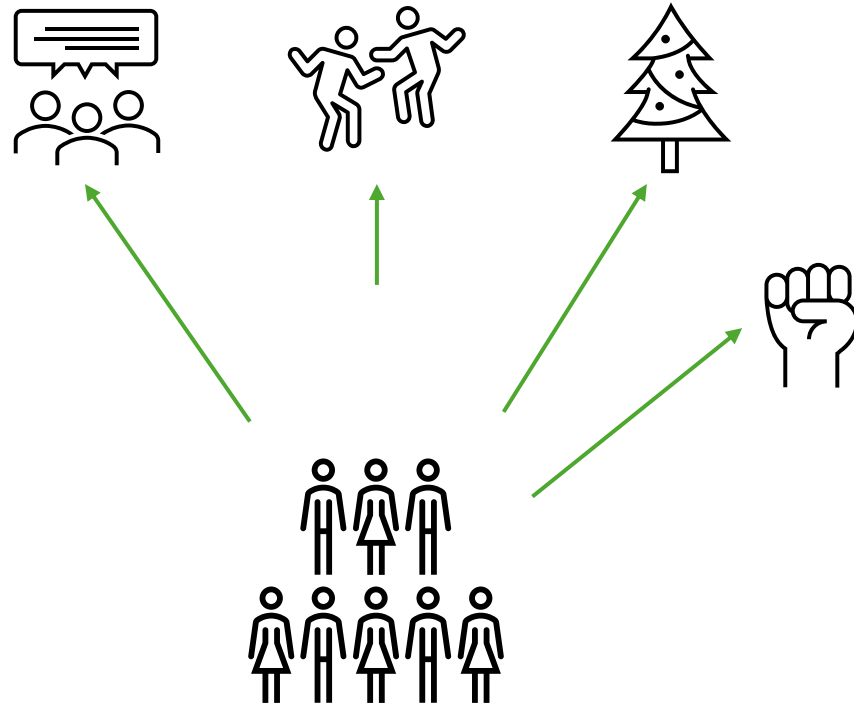


Monthly Patterns of MGEs by Type and Location

Location Indoor Outdoor Outdoor & indoor

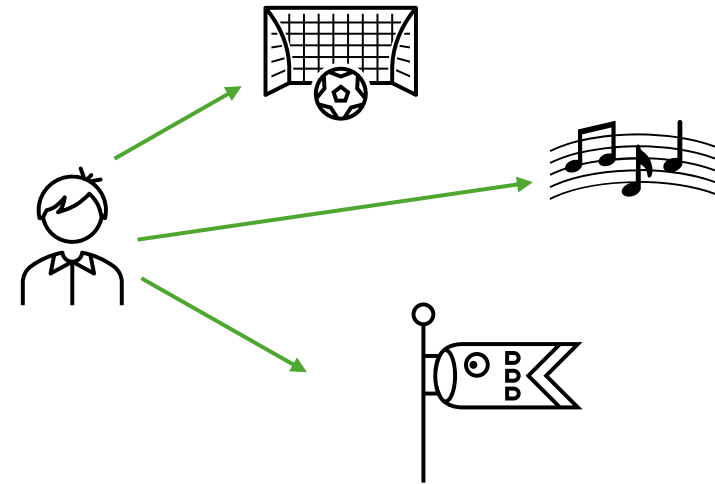


Rationale for the clustering analysis



Before

- Double-counting
- Groups are not mutually exclusive



Now

- Captures mixtures, not single types
- Groups are not mutually exclusive

Steps for clustering analysis

- ❑ To identify distinct profiles of mass-gathering event (MGE) attendance at the participant level or patterns
- ❑ 4,209 participants reported information of 10,921 events in 2024* -> 2,868 participants
- ❑ Step 1: change the data structure

```
> event %>%
+   select(id, Yes_no, Type_new)
```

```
# A tibble: 10,921 × 3
```

	id	Yes_no	Type_new
	<int>	<fct>	<fct>
1	1	Yes	Cultural event
2	1	Yes	Cultural event
3	1	Yes	Cultural event
4	1	Yes	Cultural event
5	2	Yes	Others
6	3	Yes	Cultural event
7	3	Yes	Cultural event
8	4	Yes	Professional event
9	4	Yes	Cultural event
10	5	Yes	Cultural event

```
# i 10,911 more rows
```

```
> event_part_yes_nona
```

```
# A tibble: 2,868 × 9
```

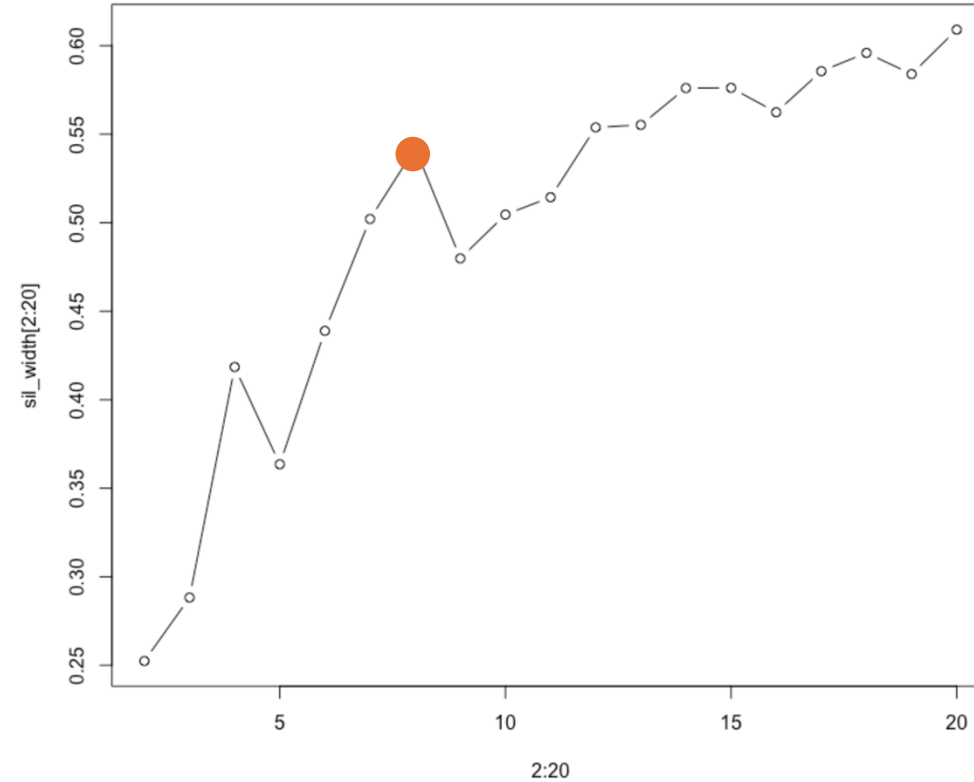
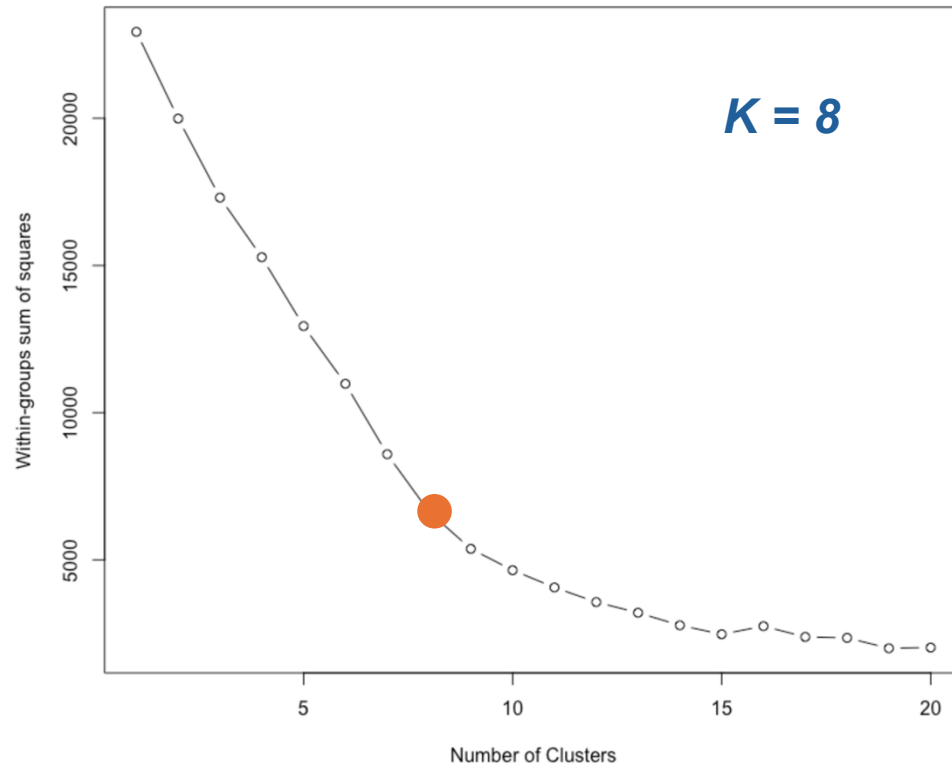
	id	n_cultural	n_sporting	n_private	n_political	n_professional	n_scientific	n_fair	n_others
	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>
1	1	4	0	0	0	0	0	0	0
2	2	0	0	0	0	0	0	0	1
3	3	2	0	0	0	0	0	0	0
4	4	1	0	0	0	1	0	0	0
5	5	4	0	0	0	0	0	0	0
6	6	1	0	0	0	0	0	0	0
7	7	0	3	0	0	1	0	0	0
8	9	3	0	0	0	0	0	0	0
9	10	1	0	0	0	0	0	0	0
10	11	4	0	0	0	0	0	0	0

```
# i 2,858 more rows
```

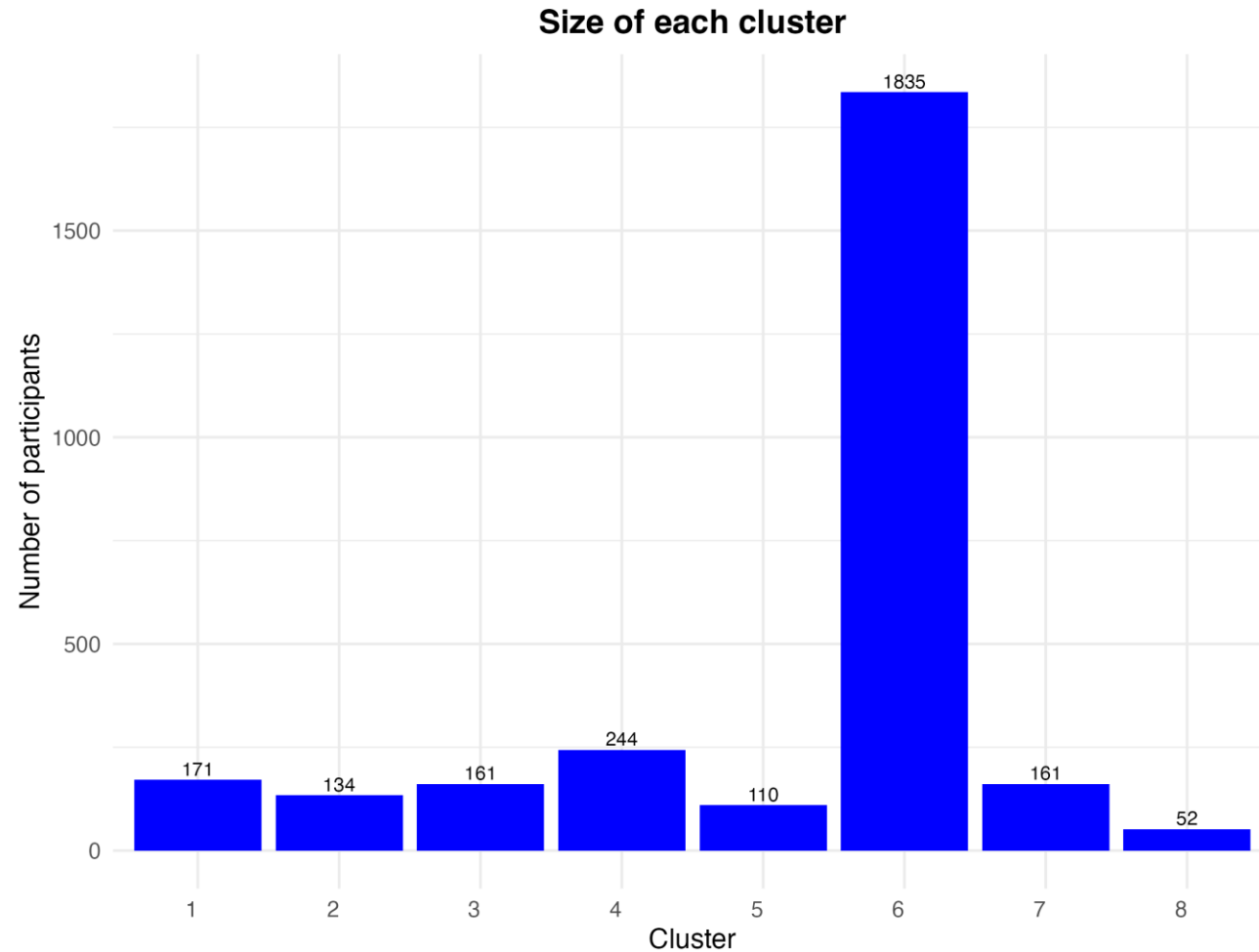
* For participants who reported attended more than 4 events, only included information of the last four events

Steps for clustering analysis

- ❑ Step 2: Computed Euclidean distance and performed hierarchical clustering
- ❑ Step 3: Applied k-means clustering for $k = 1$ to $k = 20$



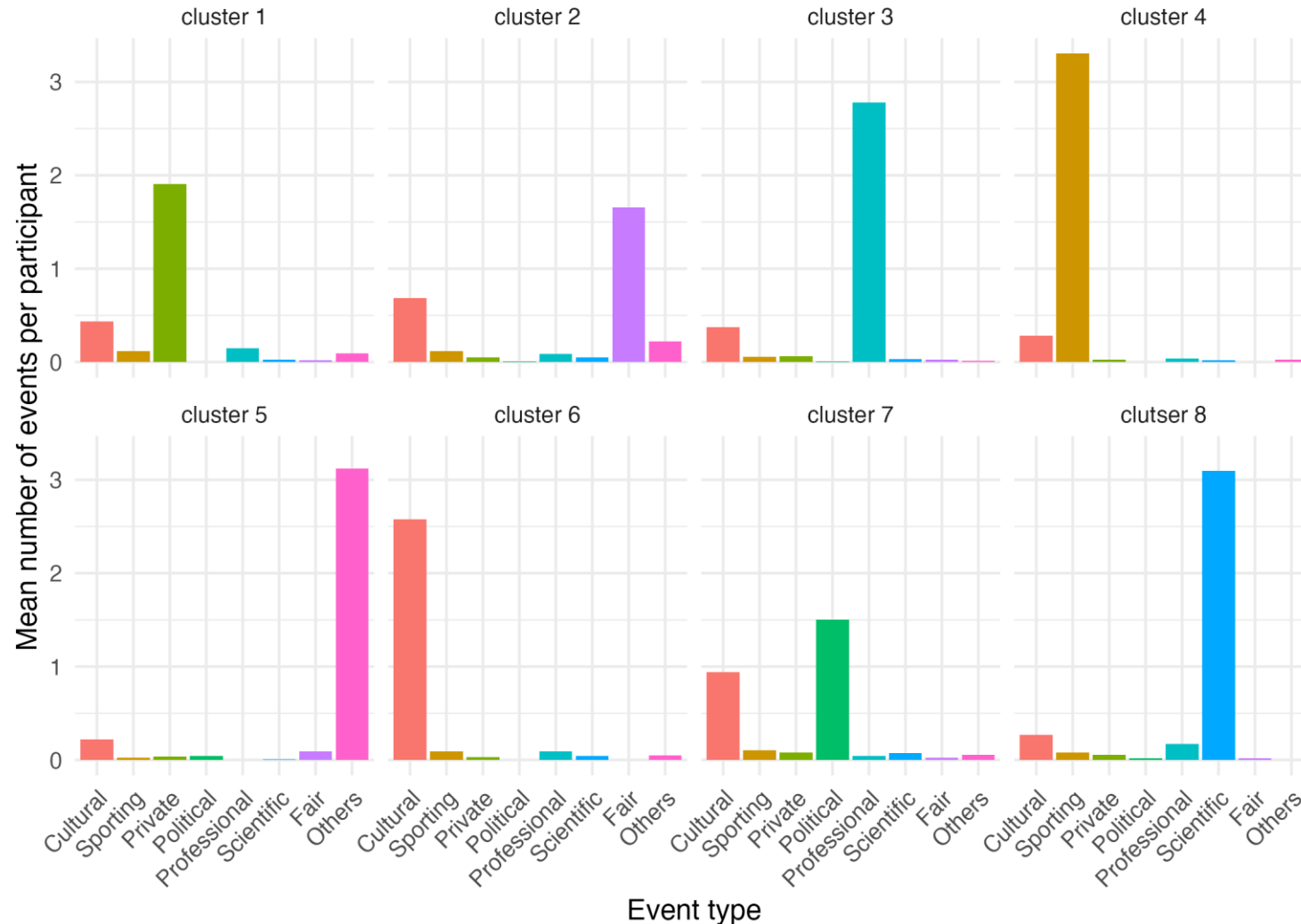
Sizes of each cluster



- Cluster 6 has the highest number of participants (1,835)
- Cluster 8 has the lowest number of participants (52)

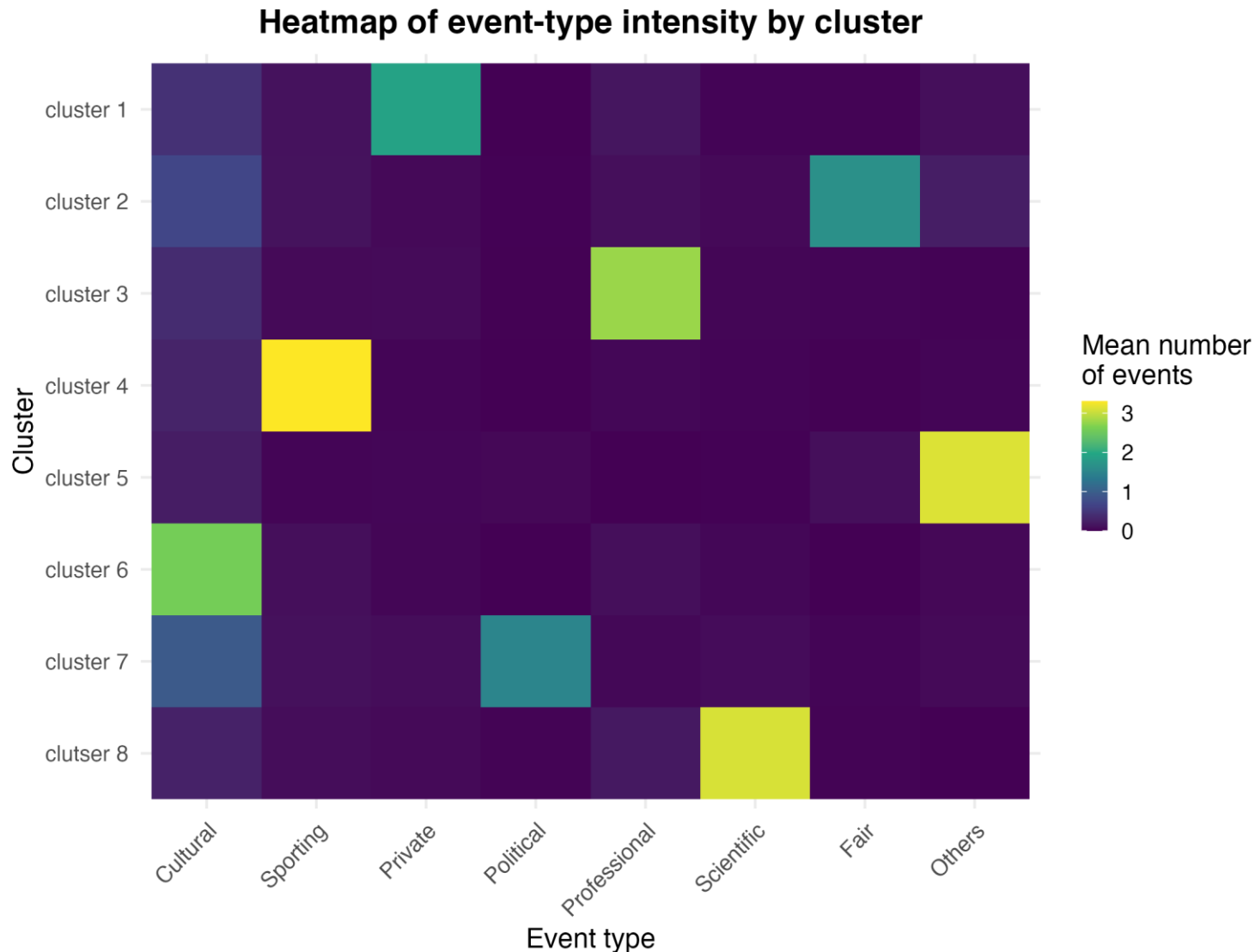
Patterns of each cluster

Event-type profiles by cluster



- Cluster 1: private, cultural
- Cluster 2: fair, cultural, others
- Cluster 3: professional, cultural
- Cluster 4: sporting, cultural
- Cluster 5: Others, cultural, fair
- Cluster 6: dominated by cultural
- Cluster 7: political, cultural
- Cluster 8: scientific, cultural, professional

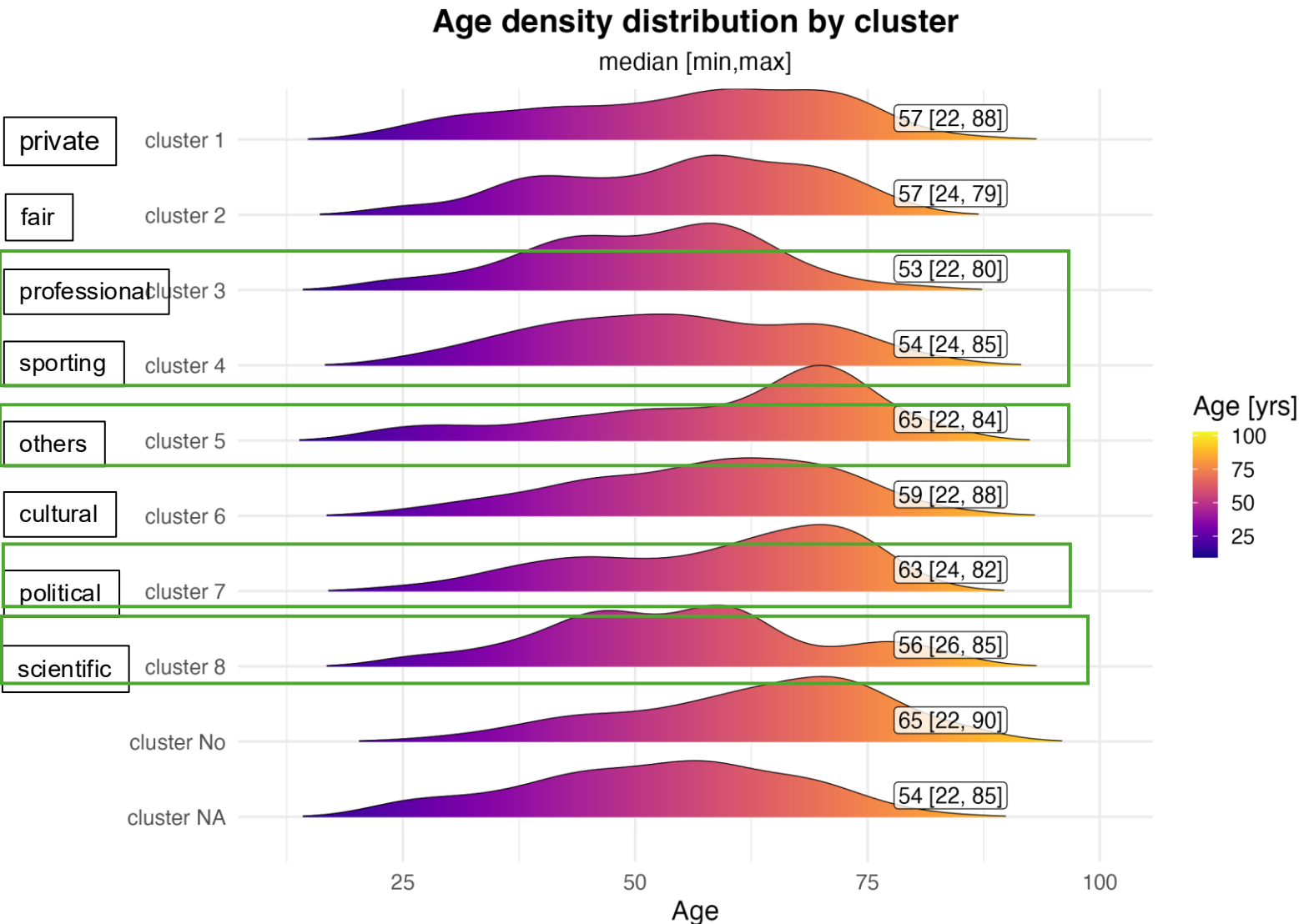
Patterns of each cluster



- Mean number of events was calculated.

$$\frac{\text{The total number of attendance one type event}}{\text{The number of participants in this cluster}}$$

Age by cluster

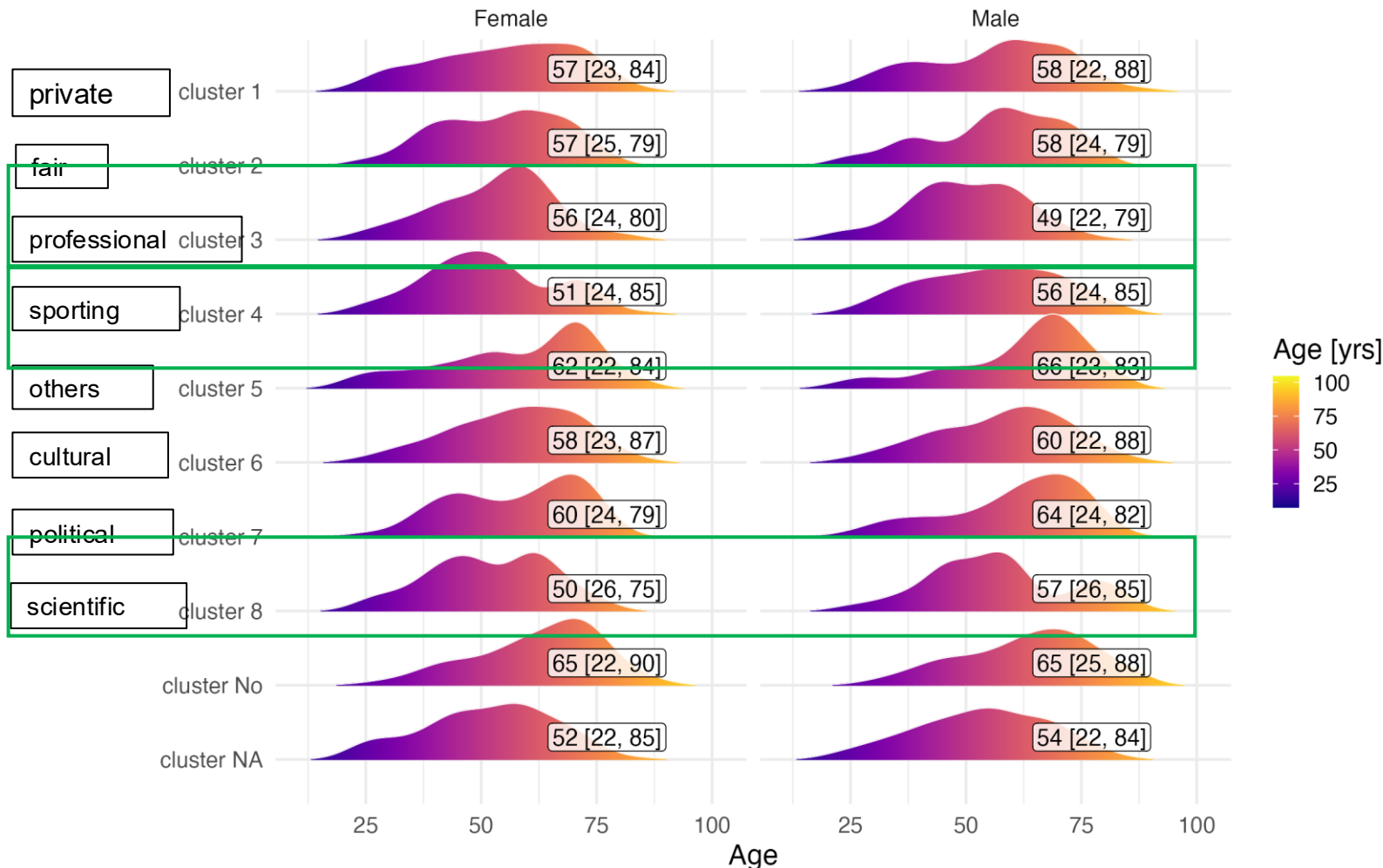


- Cluster 3 (professional), 4 (sporting), 8 (scientific) have relative younger age structure
- Cluster 5 (others), 7 (political) have older age structure
- Cluster 2 (fair), 3 (professional), 7 (political), 8 (scientific) presented two-peak shape.

Age group and sex by cluster

Age density distribution by cluster and sex

median [min,max]

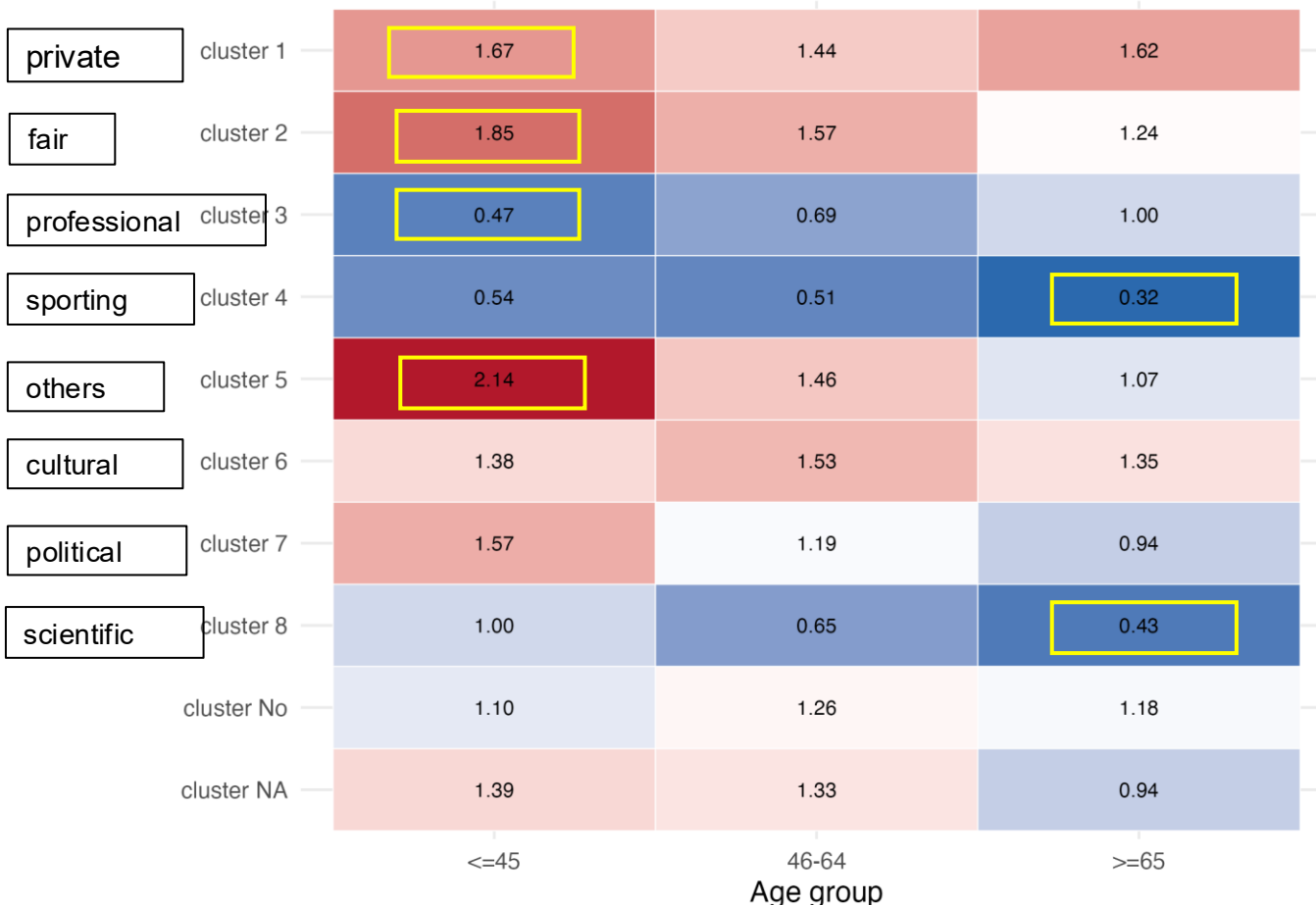


- Comparing the median age between sex, for Cluster 3 (professional, 7 yrs apart), 8 (scientific, 7 yrs apart), 4 (sporting, 5 yrs apart) there are big age difference (≥ 5 yrs)

Sex ratio by cluster

Female:male ratio by cluster, age (heatmap)

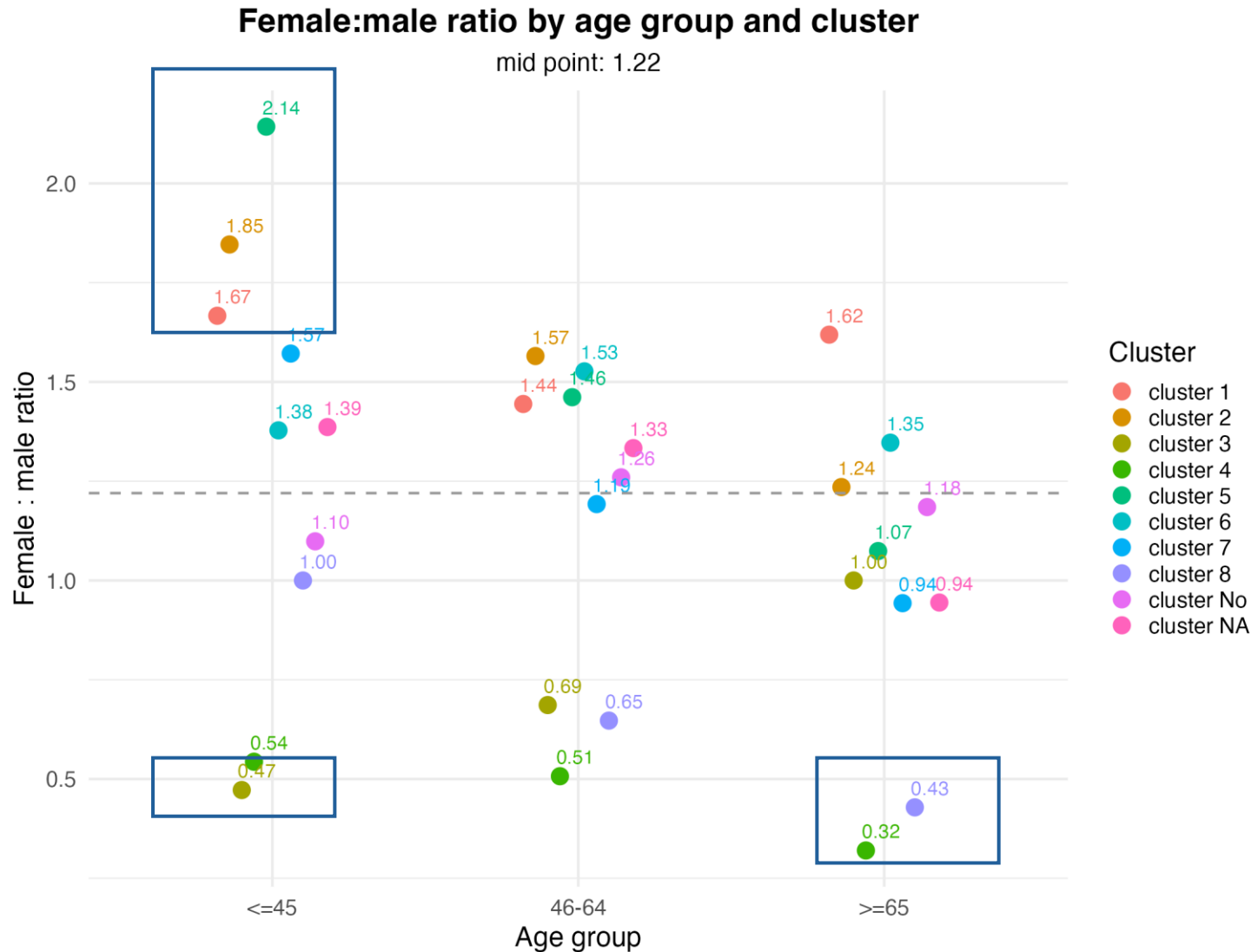
mid point: 1.22



Ratio
2.0
1.5
1.0
0.5

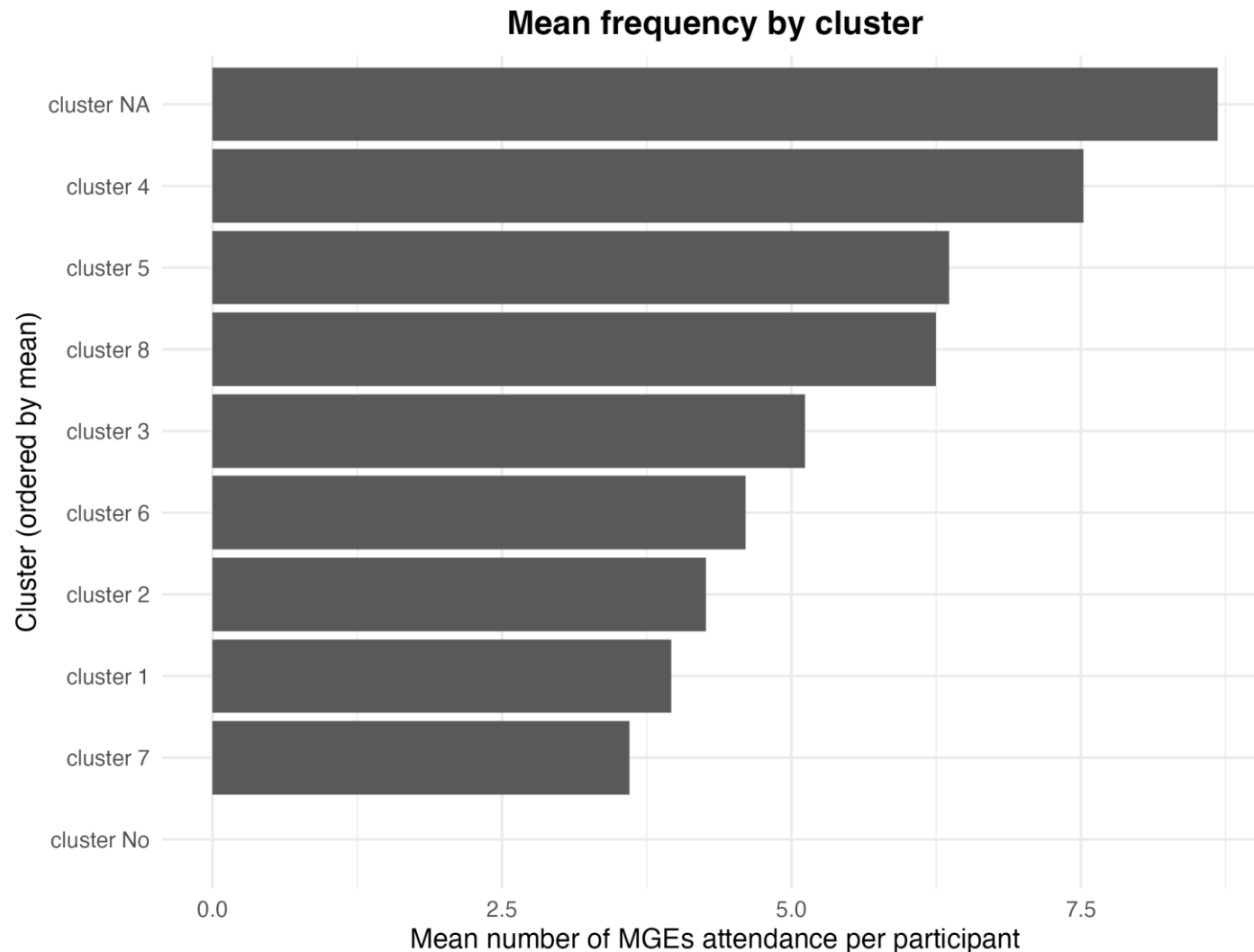
- The mid point 1.22 is the ratio of number of female participants divided by the number of male participants
- Female has higher proportion for cluster 5 (2.14, others, aged ≤ 45), cluster 2 (1.85, fair, aged ≤ 45), cluster 1 (1.67, private, aged ≤ 45).
- Male has higher proportion for cluster 4 (0.32, sporting, aged ≥ 65), cluster 8 (0.43, scientific, aged ≥ 65), and cluster 3 (0.47, professional, aged ≤ 45)

Sex ratio by cluster



- Same interpretation as previous slide

MGEs Frequency by cluster

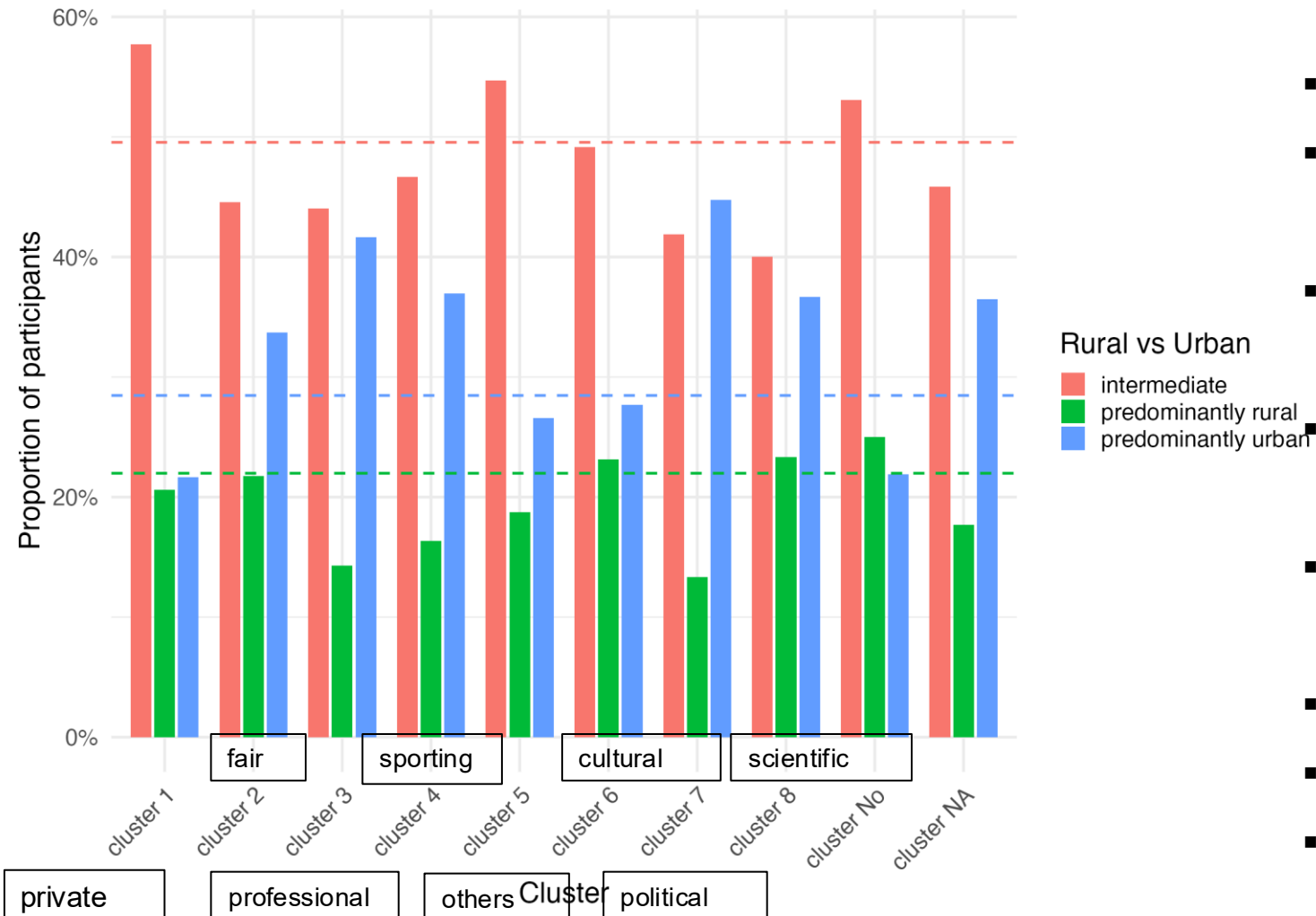


- 5-10 ~ 7, >10 ~ 12.
- Cluster 4 (dominated by sporting) has the highest mean of frequency (7.52)
- Followed by cluster 5 (6.36, dominated by others) and cluster 8 (6.25, dominated by scientific events)

Rural vs urban by cluster

Proportion of population density by cluster

---: baseline proportion



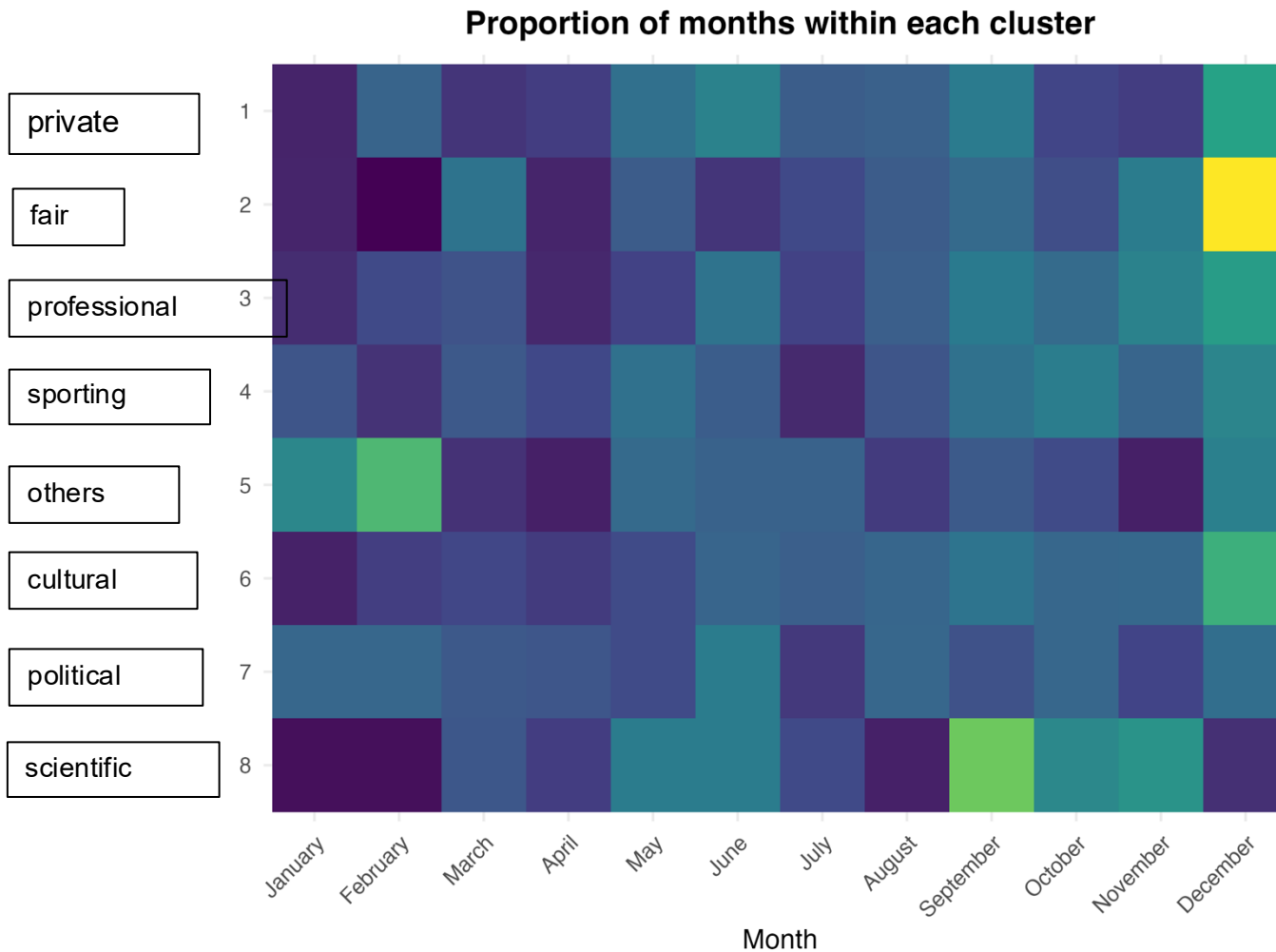
- Rural vs urban information only available for 2,779 of 4,209 participants
- Compared to the baseline proportion:
- Cluster 1 (private): higher intermediate, lower urban
- Cluster 2 (fair): higher urban, lower intermediate
- Cluster 3 (professional): much higher urban, lower rural
- Cluster 4 (sporting): much higher urban, lower rural
- Cluster 5 (others): higher intermediate
- Cluster 7 (political): much higher urban
- Cluster 8 (scientific): higher urban

Statistical result

- ❑ Outcome: infection status from October to December 2024. 0 = not infected, 1 = infected
- ❑ Model: multivariable logistic regression with logit link
- ❑ Predictors: Age, Sex, MGE cluster, Education level, Income
- ❑ Cluster 2 showed about two-fold higher odds of infection (OR $\approx$ 1.95, CI 95%; 0.97-3.95).
- ❑ Interpretation: participants from cluster 2 (dominated by fair or festival events) tended to have higher odds of infection, but this association did not reach conventional statistical significance.

<i>Predictors</i>	Infection (Yes vs No)		
	<i>Odds Ratios</i>	<i>CI</i>	<i>p</i>
(Intercept)	6.76	2.98 – 15.81	<0.01
Age	0.98	0.97 – 0.98	<0.01
Gender [Male]	0.81	0.67 – 0.98	0.03
cluster [cluster 2]	1.95	0.97 – 3.95	0.06
cluster [cluster 3]	0.88	0.45 – 1.70	0.70
cluster [cluster 4]	0.90	0.49 – 1.64	0.73
cluster [cluster 5]	1.61	0.78 – 3.34	0.20
cluster [cluster 6]	1.27	0.78 – 2.07	0.34
cluster [cluster 7]	1.27	0.65 – 2.49	0.49
cluster [cluster 8]	1.21	0.49 – 3.05	0.67
cluster [cluster No]	1.27	0.76 – 2.11	0.36
cluster [cluster NA]	1.70	0.96 – 3.00	0.07
Education [medium]	0.63	0.34 – 1.13	0.13
Education [high]	0.67	0.36 – 1.20	0.19
Income [Medium]	0.98	0.70 – 1.37	0.90
Income [High]	1.04	0.73 – 1.47	0.83
Observations	1959		
R ² Tjur	0.044		

Events patterns by cluster



- Cluster 2 (fair, cultural, others): more likely to happen in December
- Cluster 5 (others): more likely in February
- Cluster 8 (scientific): more likely in September

Events patterns by cluster

